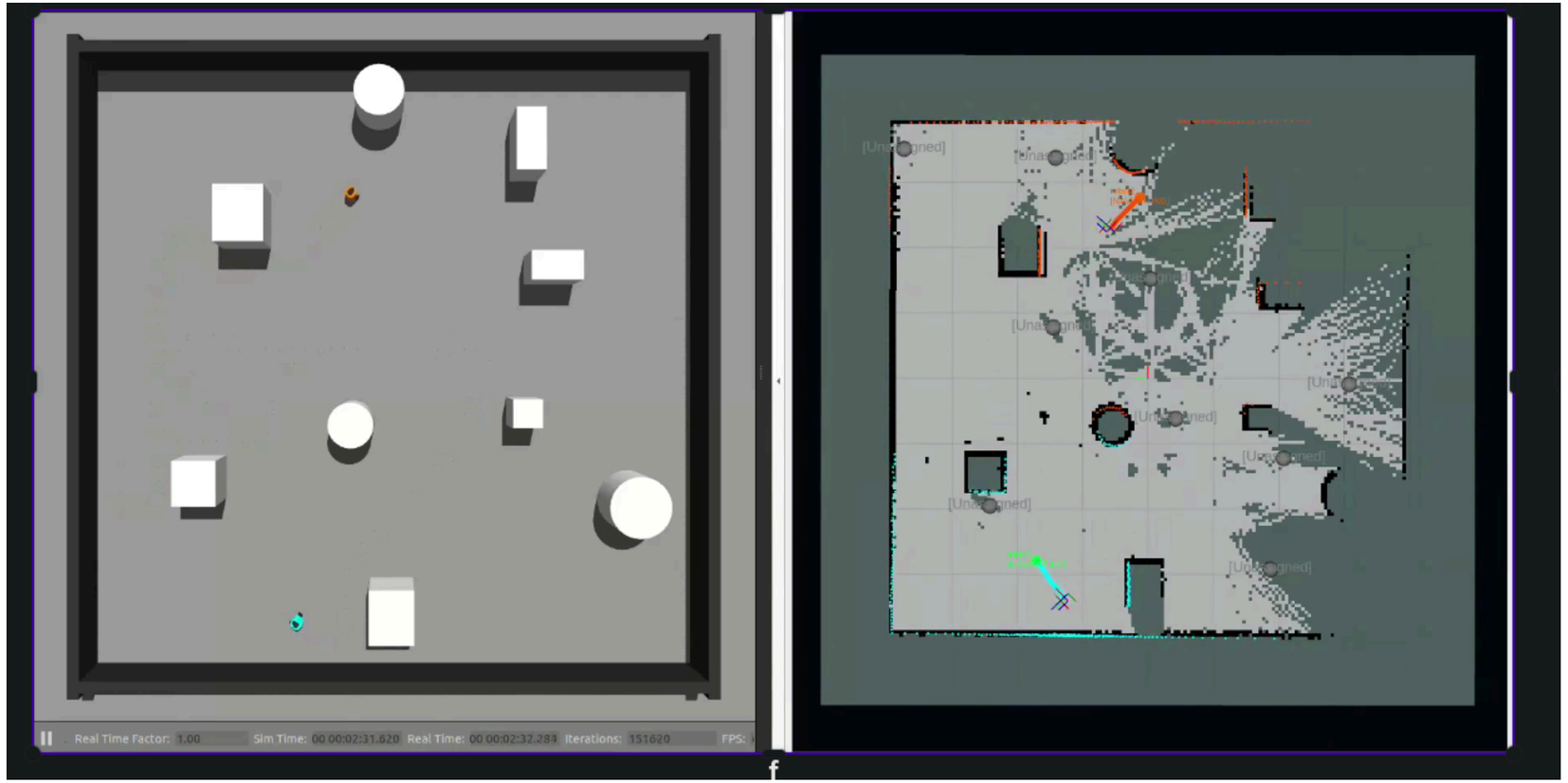
Multi-Robot Frontier Exploration with ROS 2 and Reinforcement Learning

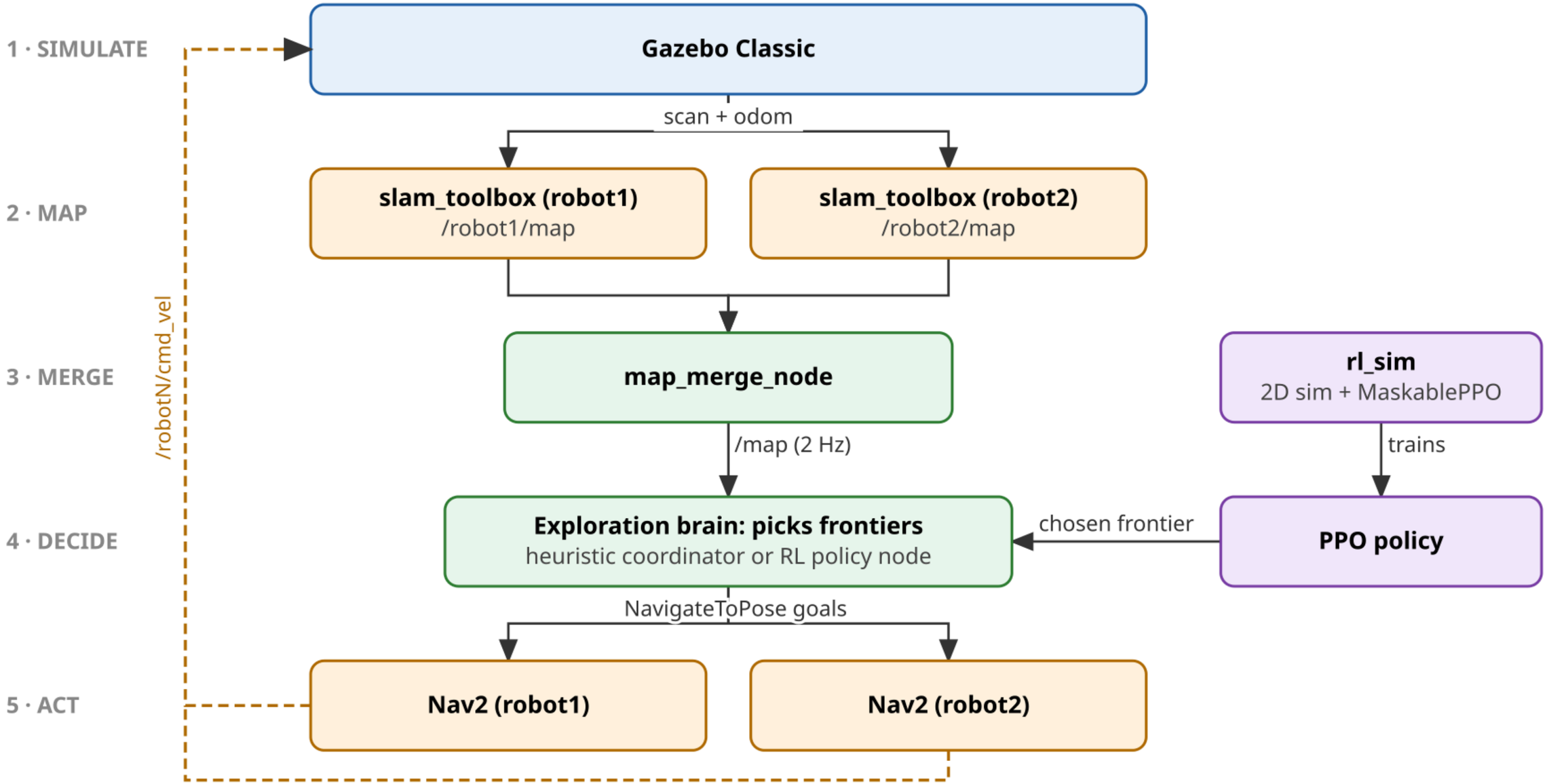
THE PROBLEM: MAP AN UNKNOWN ENVIRONMENT AS FAST AS POSSIBLE



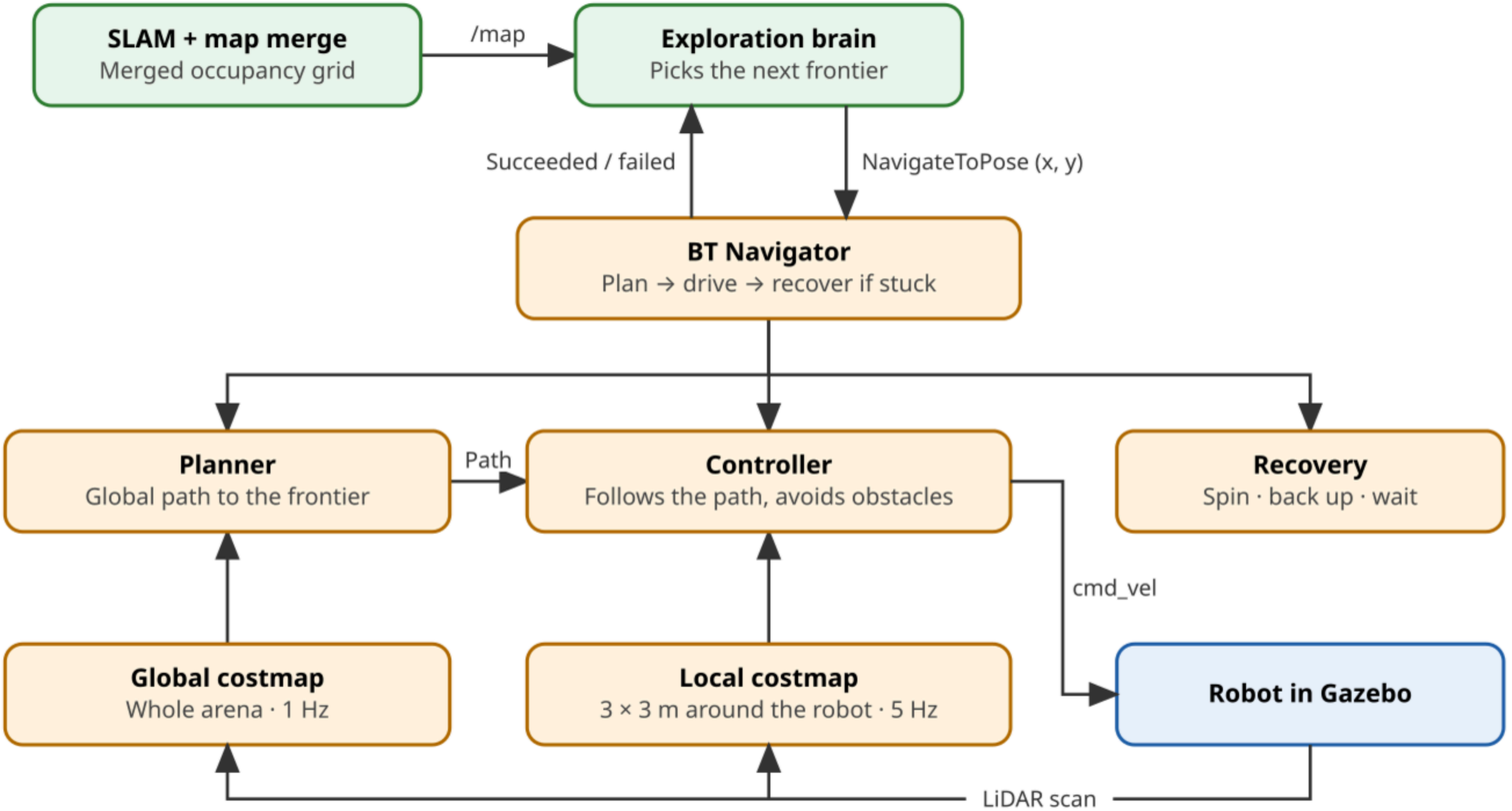
MAIN QUESTION

Can reinforcement learning perform better than a hardcoded rule for frontier based swarm exploration?

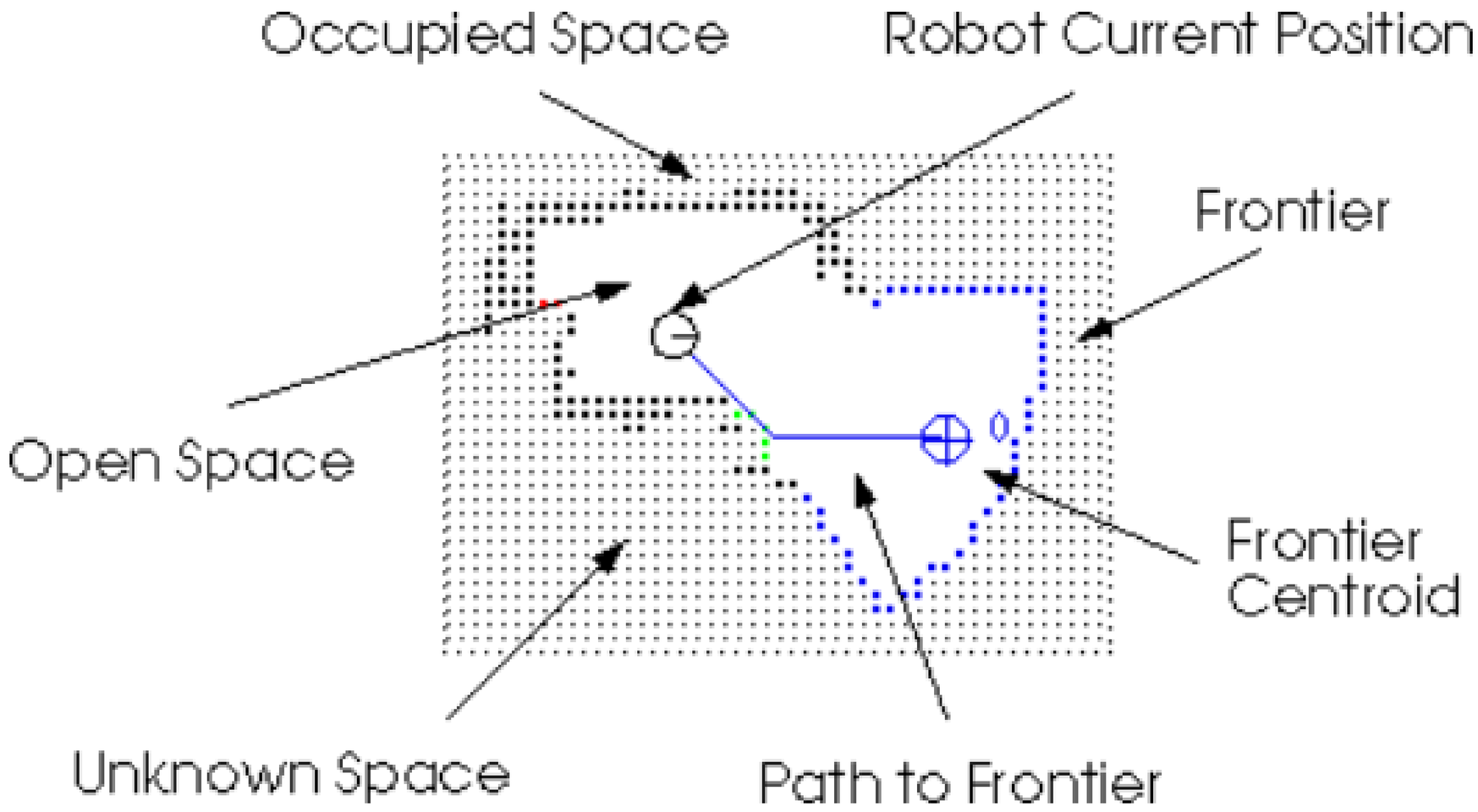
SYSTEM ARCHITECTURE: DATA AND CONTROL FLOW



NAV2 ARCHITECTURE



FRONTIER DETECTION AND EXPLORATION



HEURISTIC OVERVIEW

Prepare the Masks

↓

Detect Frontier Cells

↓

Filter Frontier Targets

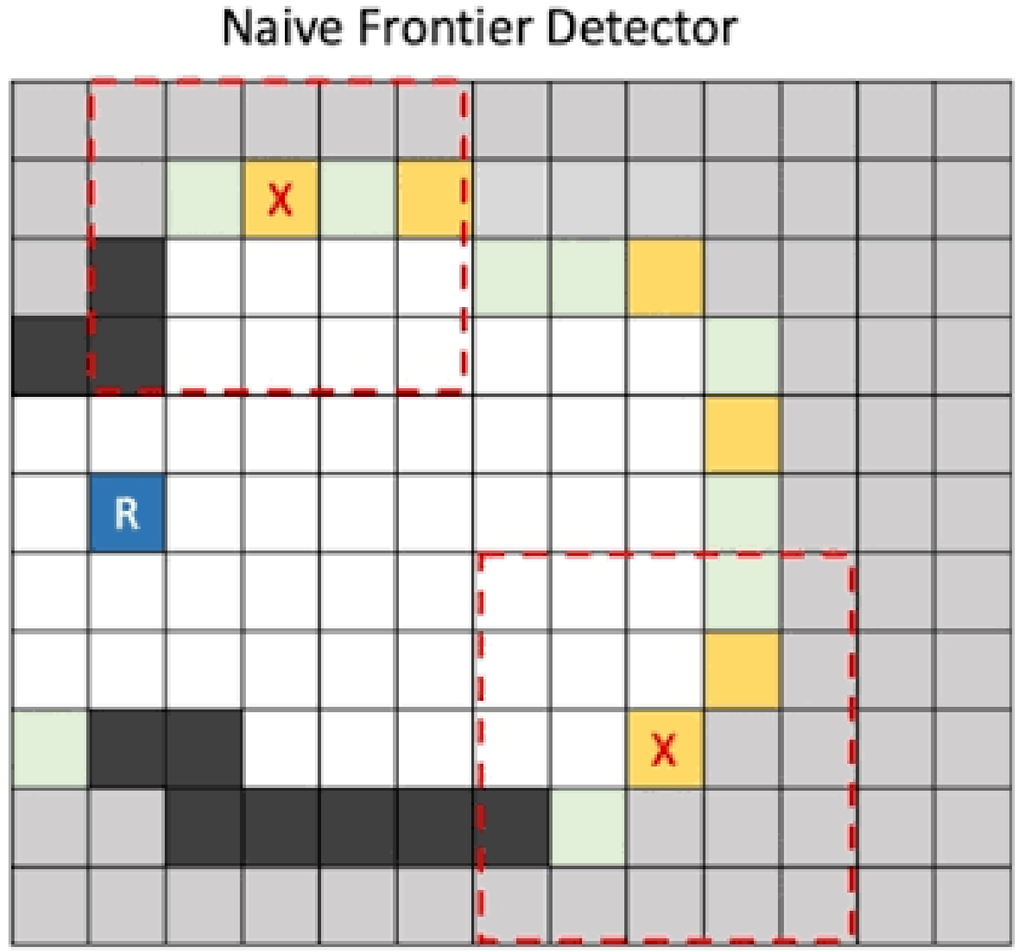
↓

Assign Targets to Robots

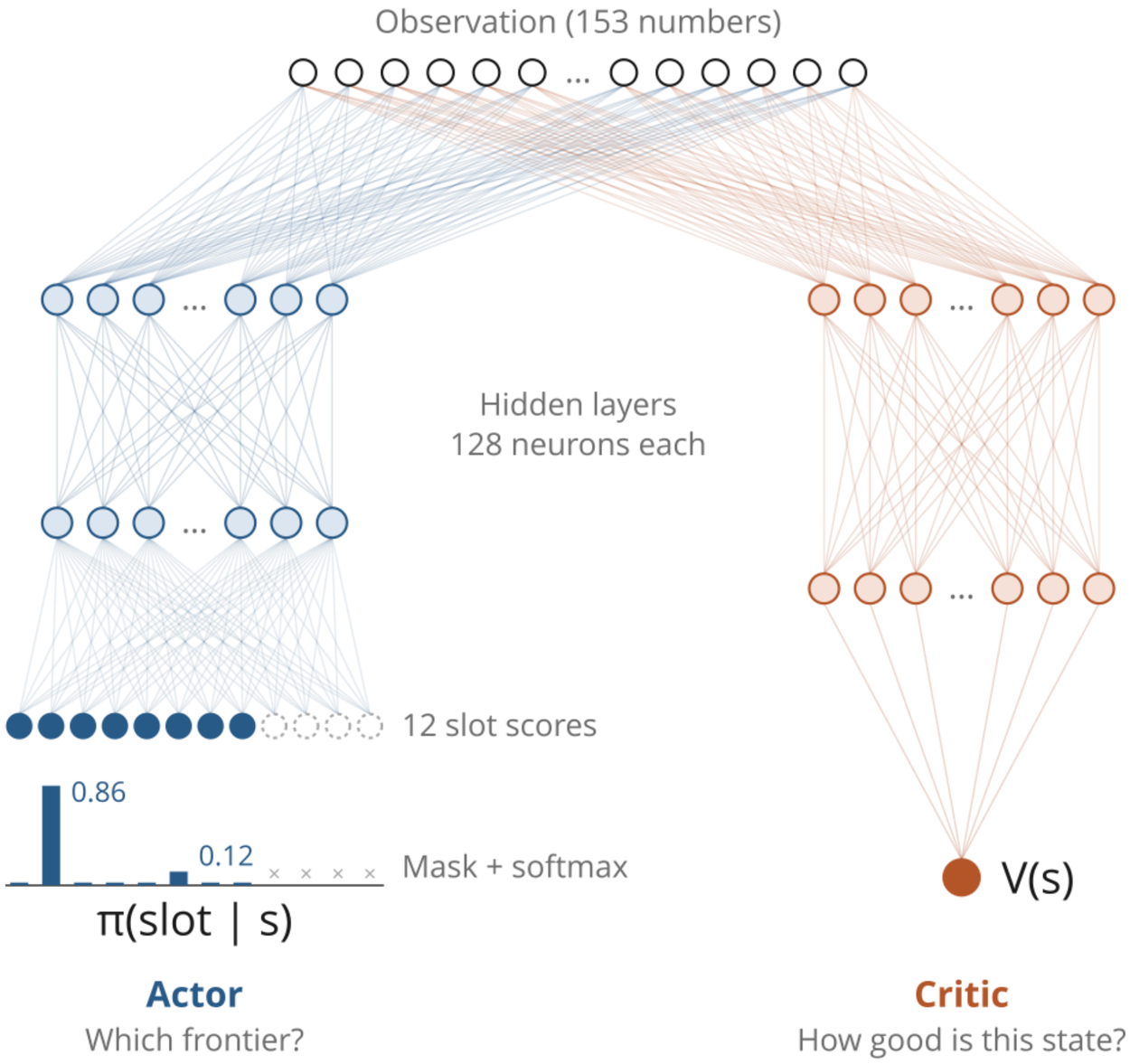
↓

Replan continuously

cost = distance + separation+ region



RL OVERVIEW



Reward

$$r = 0.1 \cdot \Delta A - 0.01 \cdot \Delta t$$

Training loop (PPO)

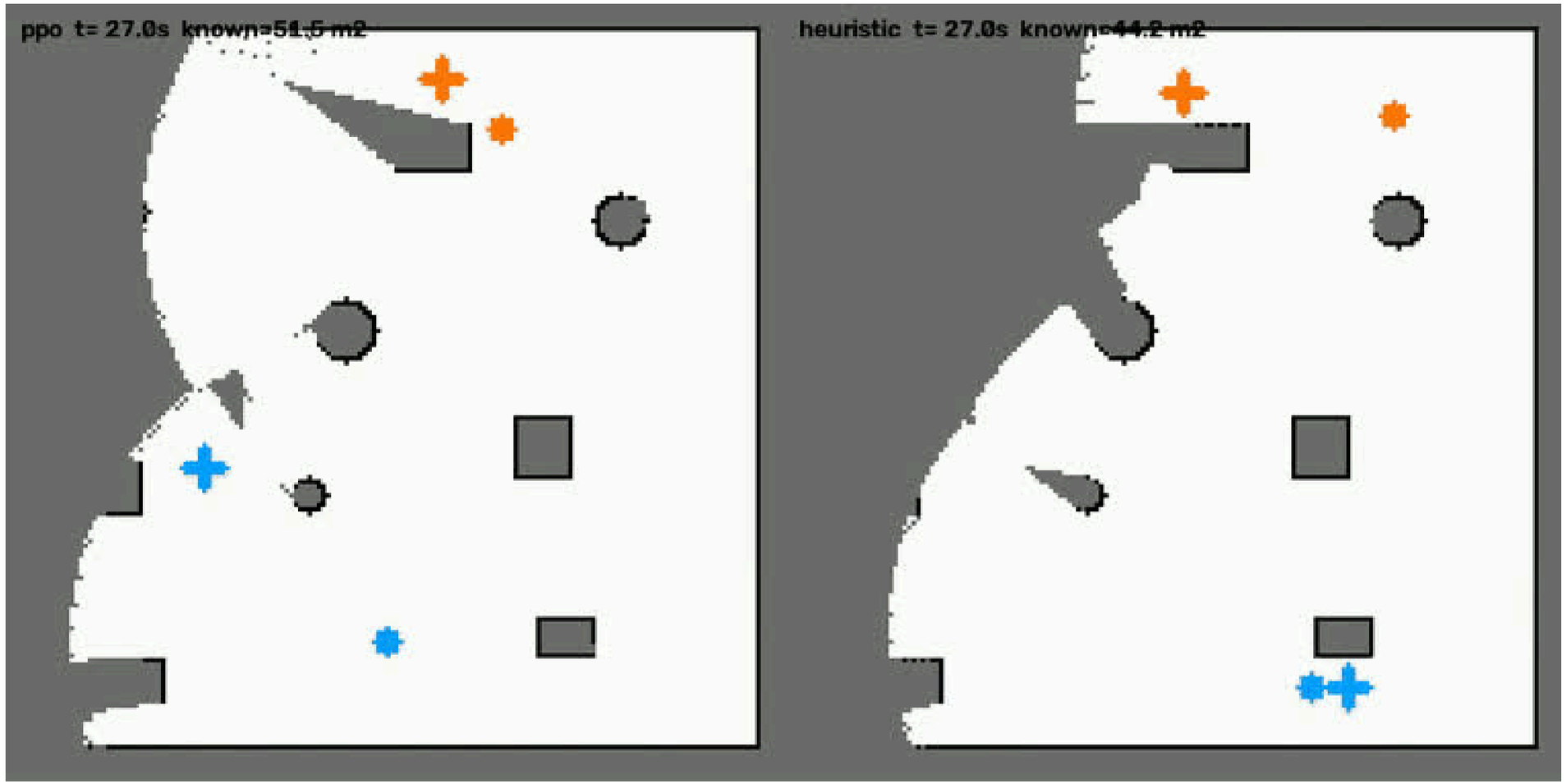
1. **Run the policy, record actions and rewards**
2. **Score each action (advantage)**
3. **Make good actions more likely (clipped)**
4. **Improve the critic's predictions**

repeat

Objective

$$J(\theta) = \underbrace{J_{CLIP}}_{\text{improve the actor}} - \underbrace{0.5 \cdot E_{VF}}_{\text{fix the critic's error}} + \underbrace{0.01 \cdot H(\pi)}_{\text{keep exploring}}$$

THE 2D TRAINING SIMULATOR



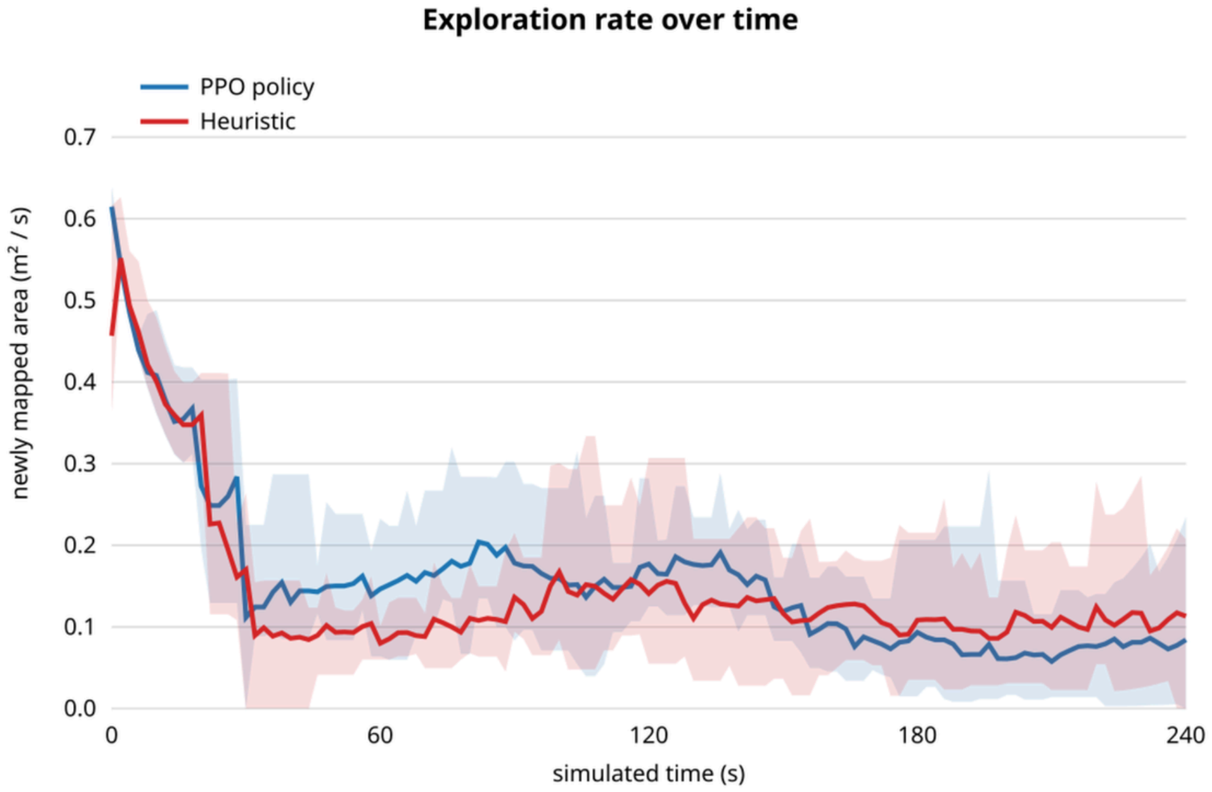
Train in a fast 2D simulator

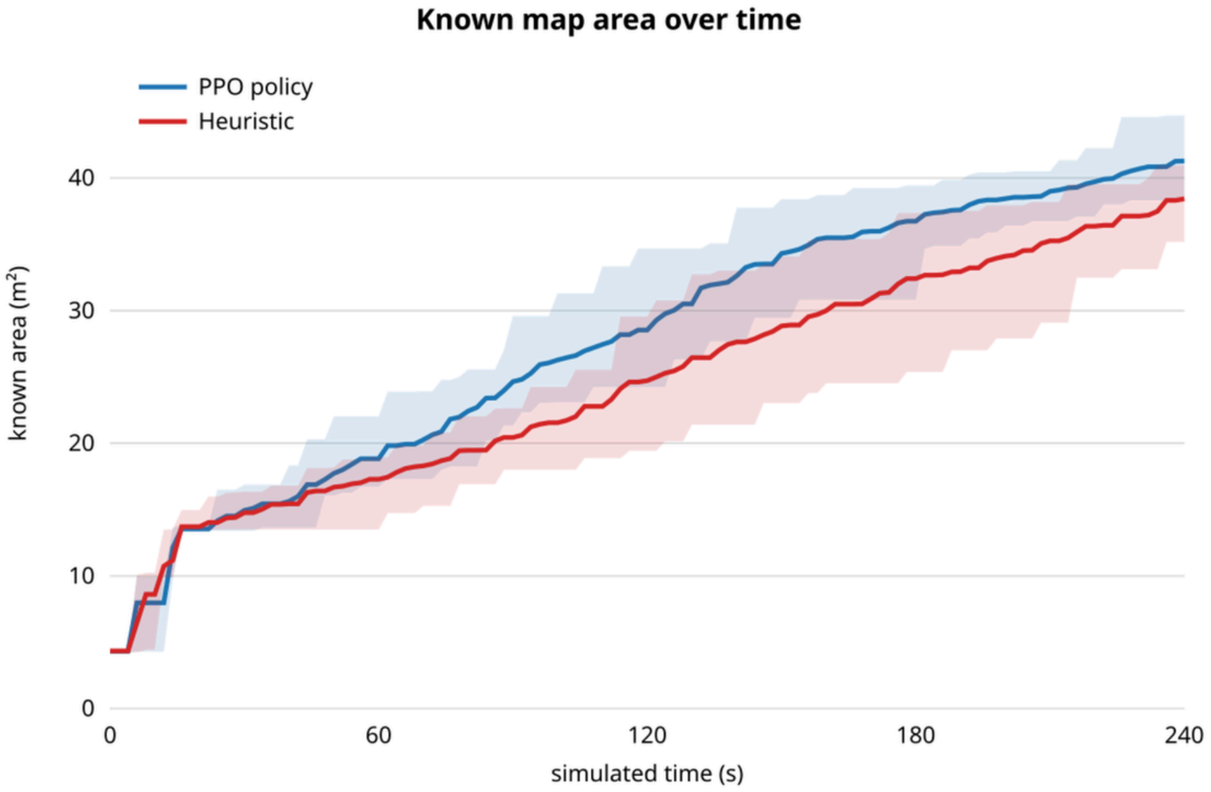
≈ 300× faster than Gazebo

190 vs 0.7 decisions / s

RL training	SB3 · MaskablePPO · PyTorch · Gymnasium · TensorBoard
2D simulator	NumPy · OpenCV

INFERENCE RESULTS





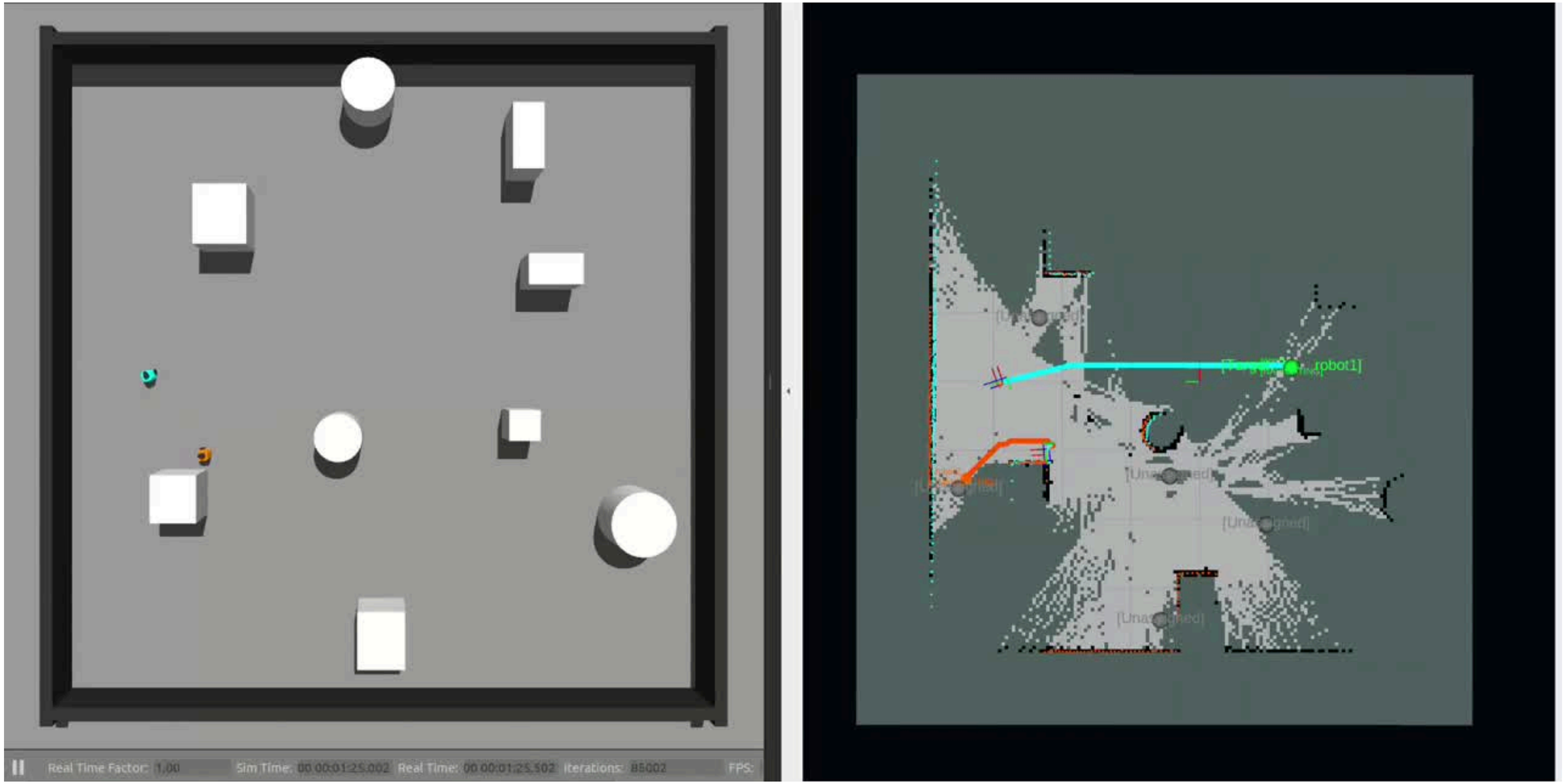
+38 %

exploration rate, 30–150 s

−21 %

time to map 30 m²

DEMO


Real Time Factor: 1.00
Sim Time: 00 00:01:25.002
Real Time: 00 00:01:25.502
Iterations: 85002
FPS: